

Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics

A Support-Visible OPH Route to the Clay Problem

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Abstract

The Clay Yang–Mills problem asks for a nontrivial four-dimensional quantum Yang–Mills theory with a positive mass gap. OPH gives a route on its compact-gauge branch.

The construction isolates the finite OPH repair mechanism and the certificate needed before that mechanism may be read as a four-dimensional Yang–Mills theorem. Fixed-cutoff local repair has a uniform rate floor and a positive transfer form. The Clay-facing statement is conditional on a renormalized continuum certificate: Schwinger-function convergence, reflection positivity, Euclidean covariance and locality, nontriviality, and Mosco or exact direct-limit convergence of the transfer/intertwiner data. Without that certificate, the finite repair branch is not by itself a construction of the four-dimensional Yang–Mills measure.

What This Paper Contributes

The standard target is clear: construct a nontrivial four-dimensional quantum Yang–Mills theory for every compact simple gauge group and prove a positive mass gap. OPH imports the usual compact gauge language, reflection positivity, and Osterwalder–Schrader/Wightman target conditions. The new ingredient is the repair interpretation of the vacuum sector.

On the support-visible compact-gauge branch, local holonomy data and the OPH scaling chart give the candidate Euclidean Yang–Mills form. Exact local repair acts as a positive Euclidean relaxation generator at finite cutoff. A uniform fixed-cutoff repair rate floor transports to the continuum Hamiltonian only when the transfer forms, vacuum projections, and finite intertwiners converge in the sense stated below. The proof therefore separates the finite repair mechanism from the additional four-dimensional continuum certificate required by the Clay problem.

1 Claim Boundary

This paper gives a proof-bearing projective weak-* / GNS extraction from the declared finite support-visible compact-gauge cylinder system. The finite positive-transfer and repair-gap mechanism is conditional on the finite ground-state-transform/cross-fiber and uniform-gap receipts. The

Clay-facing Yang–Mills theorem additionally depends on the continuum certificate stated in Assumption 9.4: renormalized four-dimensional identification, reflection positivity and OS regularity, transfer/vacuum convergence, and noncollapse.

More precisely, the imported four-dimensional form theorem uses compact-gauge reconstruction, the four-dimensional scaling chart, the reflection-positive ordinary vacuum sector, the absence of additional gauge-invariant relevant dimension-four pure-gauge operators beyond the positive quadratic curvature invariant, the support-visible cylinder extraction, and the renormalized regularity/universality receipt. The later mass-gap step adds the finite transfer receipt, exact local repair, bounded-color active collars, repair completeness, and the uniform-gap certificate.

Acceptance as a Clay-admissible solution still depends on supplying the unproved parts of the OPH compact-gauge continuum certificate: renormalized Schwinger-function convergence, reflection positivity, Euclidean covariance/locality, nontriviality, and transfer/intertwiner convergence at the strength required by the Clay/Jaffe–Witten statement. The proof below then isolates the finite repair mechanism and shows that, on any branch carrying that certificate, the gap is identified exactly:

$$\Delta_{\text{YM}} = \Delta_{\text{rep}}.$$

The two-dimensional heat-kernel identity in the wider OPH stack remains a separate normalization and worldsheet-effective bridge; it is not the mass-gap proof.

2 Position in the OPH Paper Stack

This paper is a focused companion to the OPH paper stack. The broad reconstruction program is summarized in *Observers Are All You Need* [1]. The compact technical core is *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [2], which carries the support-visible compact-gauge repair-gap theorem inside the compact paper itself. The particle branch is separate [3]. The finite repair and quotient-normal-form machinery comes from *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [4]. The regulated screen, record, and edge heat-kernel architecture is developed in *Federated Echoshedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [5].

The edge-sector theorem in the stack relates OPH heat-kernel weights to a two-dimensional Yang–Mills partition identity. That identity fixes normalization and the controlled worldsheet effective bridge. The four-dimensional mass-gap argument uses repair dynamics: exact local repair becomes a positive Euclidean relaxation generator, and the uniform repair gap is transported to the compact-gauge Hamiltonian only under the continuum certificate.

The Clay-facing import is the compact paper’s D7 Yang–Mills theorem surface: the compact-gauge reconstruction ladder, the four-dimensional Euclidean Yang–Mills form theorem, the coherent compact-gauge extraction proposition, the conditional support-visible Osterwalder–Schrader reconstruction theorem, the nontriviality certificate, and the exact finite repair-gap theorem. This note isolates that branch theorem surface; it does not enlarge it.

3 The Clay Target

The Clay Mathematics Institute describes the Yang–Mills mass gap as the missing mathematical foundation behind the quantum theory used for nonabelian gauge forces [7]. Jaffe and Witten state the problem as follows: for every compact simple gauge group G , construct a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ satisfying axiomatic properties at least as strong as the stated Wightman or Osterwalder–Schrader references and prove a mass gap $\Delta > 0$ [8, 9].

The OPH proof below addresses that target through a different variable. The starting data are finite observer patches, compact-gauge visible quotient data, and exact repair collars. The nonzero energy threshold is the cost of leaving the repair-fixed vacuum sector.

4 Standing Setup

Fix a compact simple gauge group G carried by an OPH compact-gauge zero-obstruction vacuum branch, realized at fixed cutoff by the declared compact-gauge patch-carrier architecture. The architecture supplies finite local Hilbert spaces, exact local constraints, patch and overlap algebras, overlap sector projectors, record layers, and local repair interfaces.

Let r range over a cofinal refinement family of finite regulators. For each r , let

$$(\mathcal{H}_r, \Omega_r, H_r), \quad T_r(t) = e^{-tH_r},$$

be the physical Euclidean Hilbert space, vacuum, Hamiltonian, and transfer semigroup.

Let $X_r := X_r^{(0)}$ be the support-visible time-zero compact-gauge quotient configuration space, let π_r be the stationary time-zero measure, and set

$$K_r := L^2(X_r, \pi_r).$$

Write $\omega_r(a) := \int_{X_r} a d\pi_r$ for the associated state functional; K_r already implements the finite π_r -null support reduction. Let $\mathcal{C}_r$ be the finite family of active repair collars. For each active collar $C \in \mathcal{C}_r$, let

$$\rho_C := \rho_{C,r} : X_r \rightarrow Y_{C,r}$$

be the complete repaired visible datum, and let

$$E_C : K_r \rightarrow K_r$$

be conditional expectation onto the ρ_C -measurable functions.

The proof stays on the ordinary or central zero-obstruction vacuum branch. The genuinely noncentral higher-gauge branch is a different fixed-cutoff sector in the OPH stack and is not used for the ordinary compact-simple G theorem below.

Notation	Meaning
X_r, π_r, K_r	support-visible time-zero quotient configuration space, stationary measure, and $L^2(X_r, \pi_r)$.
$\mathcal{C}_r$	finite active repair-collar family at regulator r .

Notation	Meaning
ρ_C	complete repaired visible datum on collar C .
E_C	π_r -preserving conditional expectation onto ρ_C -measurable functions.
L_r^{rep}	ground-state transformed Euclidean repair generator at cutoff r .
$P_{0,r}$	projection onto constants in K_r , corresponding to the physical vacuum.
U_r	finite-stage unitary from the physical Euclidean Hilbert space to the repair L^2 space.
$K, \mathcal{H}, U$	support-visible continuum repair Hilbert space, physical Hilbert space, and limiting unitary.
c_*	uniform active-collar repair-rate floor.

5 Imported 4D Euclidean Yang–Mills Form

Assumption 5.1 (Support-visible compact-gauge Yang–Mills branch). *The compact-gauge branch used below satisfies the following branch-local conditions.*

- (i) *the ordinary or central zero-obstruction compact-gauge sector survives refinement with compact simple structure group G ;*
- (ii) *the support-visible quotient carries a four-dimensional Euclidean scaling chart;*
- (iii) *the ordinary vacuum sector is reflection positive and has topological angle $\theta = 0$;*
- (iv) *the local finite-constraint MaxEnt/Gibbs family is gauge-invariant, Euclidean local, rotation-invariant, and refinement-stable;*
- (v) *no additional gauge-invariant relevant dimension-four pure-gauge operator remains on this branch besides the positive quadratic curvature invariant;*
- (vi) *the branch is carried on a separated cofinal regulator system with the finite compact-gauge cylinder data defined in Section 9.*

No continuum cylinder state, GNS space, transfer generator, or vacuum projection is assumed in this branch declaration.

Theorem 5.2 (Imported conditional OPH four-dimensional Euclidean Yang–Mills form). *Under Assumptions 5.1 and 9.4, on an extracted cylinder family from Proposition 9.2 that lies on the declared local four-dimensional scaling branch, the continuum gauge-sector Euclidean action is*

$$S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle d^4x, \quad F = dA + A \wedge A, \quad (\text{YM})$$

with compact simple structure group G . The extracted gauge-quotient cylinder family is denoted

$$d\mu_{\text{YM}}(A) = Z^{-1} e^{-S_E[A]} DA/G$$

in the OPH support-visible GNS representation. This notation abbreviates the certified cylinder family; it is not a literal infinite-dimensional Lebesgue measure. Its support-visible continuum transfer semigroup is the corresponding Euclidean Yang–Mills semigroup.

Proof. This is the support-visible compact-gauge Yang–Mills form theorem of the compact OPH paper, where it appears as Theorem `thm:oph-4d-euclidean-yang-mills-form` [2]. The proof spine is recalled here because it fixes the target Hamiltonian for the mass-gap step.

The compact-gauge branch reconstructs a compact group G from the zero-obstruction transportable bosonic sector category and its faithful fiber functor [2]. At fixed cutoff, the declared compact-gauge patch-carrier presentation gives support-visible link holonomies and plaquette holonomies. In the refinement limit, the zero-obstruction gluing law makes infinitesimal rectangle holonomies multiplicative and path-local. The Cauchy/regularity certificate in items (Y1)–(Y3) upgrades the generalized-holonomy cluster state to a local connection A on the four-dimensional scaling chart, and the infinitesimal plaquette defect is

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \quad F = dA + A \wedge A.$$

The Euclideanized MaxEnt/local-Gibbs branch gives a local finite-range action density built from support-visible gauge-invariant collar data. Gauge quotienting permits only class functions of the curvature and its covariant derivatives. The four-dimensional scaling chart, Euclidean rotation invariance, locality, and reflection positivity leave one relevant dimension-four positive quadratic invariant in the pure gauge sector:

$$\langle F_{\mu\nu}, F_{\mu\nu} \rangle.$$

The possible topological density $\langle F \wedge F \rangle$ is reflection odd and belongs to a separate topological-angle sector; it is absent on the ordinary reflection-positive zero-obstruction vacuum branch used here. Higher curvature powers and covariant-derivative terms are irrelevant operators under the declared continuum scaling; their disappearance in the extracted state is part of the uniform renormalized remainder control in (Y1)–(Y3). Normalizing the unique positive quadratic invariant defines the coupling g and gives (YM).

The finite-stage cylinder measures are the gauge-register / quantum-link Gibbs measures pushed to the support-visible quotient. Proposition 9.2 proves the compatible support-visible weak-* / GNS cylinder extraction. The generator and transfer-semigroup identification use the separate dynamic items (Y4)–(Y5) of Assumption 9.4; weak-* compactness alone does not provide them. $\square$

Corollary 5.3 (Yang–Mills equations on the compact-gauge branch). *With an external conserved gauge current J , the OPH curvature-square action on the same support-visible compact-gauge branch gives*

$$DF = 0, \quad D * F = g^2 * J,$$

equivalently

$$D_\mu F^{\mu\nu} = g^2 J^\nu$$

in local coordinates after the usual continuation to the Lorentzian field-equation convention.

Proof. The curvature $F = dA + A \wedge A$ obeys the covariant Bianchi identity $DF = 0$. Varying the quadratic curvature action

$$S[A, J] = -\frac{1}{2g^2} \int \langle F \wedge *F \rangle + \int \langle A \wedge *J \rangle$$

with respect to compactly supported variations of A gives

$$\delta S = \int \langle \delta A \wedge (g^{-2} D^* F - *J) \rangle$$

up to a boundary term. Stationarity gives $D^* F = g^2 * J$. The source-free equation is the case $J = 0$. $\square$

Remark 5.4 (Why this step matters for the prize). The proof has two separate claims. Theorem 5.2 identifies the support-visible continuum gauge sector with four-dimensional Euclidean Yang–Mills under its branch hypotheses. The spectral argument applies to that Hamiltonian and proves a positive gap only when Assumption 9.4 supplies the OS/transfer/nontriviality bridge.

Remark 5.5 (Abelian boundary and Maxwell branch). The ordinary electromagnetic $U(1)_Q$ branch is the abelian boundary case of the same compact-gauge curvature package. Restricting $F = dA + A \wedge A$ to $\mathfrak{u}(1)_Q$ gives

$$F_Q = dA_Q, \quad dF_Q = 0, \quad d^* F_Q = g_Q^2 * J_Q$$

after varying the quadratic electromagnetic action with current J_Q . This recovers Maxwell’s equations after canonical electromagnetic normalization. The Clay-facing mass-gap claim concerns compact simple nonabelian G on the support-visible branch above. The abelian photon branch is gapless because the unbroken $U(1)_Q$ gauge symmetry protects the massless carrier.

6 Fixed-Cutoff Repair Equals Projection

Proposition 6.1 (Local exact repair equals conditional expectation). *For each active collar C , the exact-Markov repair map on the support-visible quotient is the π_r -preserving conditional expectation E_C .*

Proof. On the exact-Markov branch, repair preserves exactly the repaired visible datum ρ_C , changes only complementary invisible fiber data, and acts on the quotient-first physical algebra rather than on microscopic representatives. Let Φ_C be the Heisenberg repair map and let $\mathcal{N}_C$ be the repaired local fixed algebra. Exact repair semantics require $\mathcal{N}_C$ to be exactly the ρ_C -measurable subalgebra. They give

$$\begin{aligned} \Phi_C(a) &= a \quad (a \in \mathcal{N}_C), \\ \Phi_C(\mathcal{A}_r^{G,sv}) &\subseteq \mathcal{N}_C, \quad \omega_r \circ \Phi_C = \omega_r, \end{aligned}$$

and Φ_C is $\mathcal{N}_C$ -bimodular. Therefore, for $a \in \mathcal{N}_C$ and $x \in \mathcal{A}_r^{G,sv}$,

$$\omega_r(a^* \Phi_C(x)) = \omega_r(a^* x).$$

Since $\Phi_C(x) \in \mathcal{N}_C$, this equation characterizes its class in K_r as the orthogonal projection of x onto the ρ_C -measurable subspace, uniquely modulo π_r -null functions. That orthogonal projection is the π_r -preserving conditional expectation E_C . $\square$

7 Exact Euclidean-Consensus Law

Lemma 7.1 (Fiber-homogeneous orbit condition). *Fix an active collar C and a repaired value $y \in Y_{C,r}$. On the support-visible quotient, let the complete conditional fiber be*

$$F_C(y) = \rho_{C,r}^{-1}(y)$$

with conditional law $\nu_{C,y}$. If this is either finite uniform or standard atomless, has no remaining observable labels, and the primitive relaxation is invariant under every $\nu_{C,y}$ -preserving automorphism, then the full fiber, including every holonomy coordinate actually resampled by repair, is homogeneous for the local repair receipt.

Proof. This is the complete-fiber homogeneity receipt. Quotienting removes declared implementation labels, but the proof must also check that no remaining source constraint or relaxation rate distinguishes points in the fiber. Under that check, the conditioned state and primitive relaxation carry the stated full measure-preserving symmetry. $\square$

Lemma 7.2 (Scalar relaxation on a homogeneous conditional fiber). *Let (F, ν) be either a finite uniform probability space or a standard atomless probability space. Let E_F be expectation onto constants, and let D_F be a bounded positive self-adjoint Markov relaxation generator such that*

$$\ker D_F = \text{Ran}(E_F)$$

and D_F commutes with every measure-preserving automorphism of (F, ν) . Then there is a scalar $c_F > 0$ such that

$$D_F = c_F(I - E_F).$$

Proof. For a finite uniform fiber, the symmetric-group representation on the mean-zero subspace is irreducible. For a standard atomless fiber, use equal-measure finite partitions. Automorphisms within pieces and permutations of pieces force every bounded commutant operator to be scalar on the mean-zero step functions; compatibility under refinements gives one scalar, and these step functions are dense in $L_0^2(F, \nu)$. Positivity and the kernel condition make the scalar strictly positive. The formula follows. $\square$

Assumption 7.3 (Finite ground-state-transform and cross-fiber receipt). *For every regulator, the reflection-positive finite transfer matrix supplies a unitary $U_r : \mathcal{H}_r \rightarrow K_r$, with $U_r \Omega_r = \mathbf{1}_r$, such that*

$$U_r H_r U_r^{-1} = \sum_{C \in \mathcal{C}_r} D_C.$$

Each D_C is a bounded positive self-adjoint detailed-balance Markov generator acting on the complete complementary conditional fiber, including every record or holonomy coordinate changed by repair; its fixed algebra is exactly the $\rho_{C,r}$ -measurable algebra, and its fiberwise scalar $c_C(y)$ is independent of the repaired value y on the support of π_r . A Gibbs state by itself does not imply this finite transfer-matrix/decomposition receipt.

Theorem 7.4 (Exact Euclidean repair law under homogeneous fibers). *Under the fiber-homogeneous orbit condition for every active collar and Assumption 7.3, there are positive constants $c_C > 0$ such that the ground-state transformed physical Euclidean generator is exactly*

$$L_r^{\text{rep}} = \sum_{C \in \mathcal{C}_r} c_C(I - E_C), \tag{1}$$

and therefore

$$U_r e^{-tH_r} U_r^{-1} = e^{-tL_r^{\text{rep}}} \quad (t \geq 0), \quad (2)$$

for a unitary $U_r : \mathcal{H}_r \rightarrow K_r$ with $U_r \Omega_r = \mathbf{1}_r$.

Proof. The finite receipt supplies the ground-state transform and its primitive local pieces. Each D_C is supported on one collar C , preserves exactly the repaired visible datum $\rho_{C,r}$, and relaxes the complete complementary conditional fiber. Hence

$$\ker D_C = \text{Ran}(E_C).$$

Lemma 7.1 gives the full hidden-fiber permutation symmetry. Applying Lemma 7.2 fiberwise gives $D_{C,y} = c_C(y)(I - E_{C,y})$. The cross-fiber receipt makes $c_C(y) = c_C$, so

$$D_C = c_C(I - E_C).$$

The scalar is positive because C is active. Branch homogeneity makes it a collar-type scalar, so summing over the active collars gives (1), and exponentiation of the positive self-adjoint generator gives (2). $\square$

Lemma 7.5 (Uniform active-collar rate floor). *Assume finite local combinatorial type, a branch-homogeneous local constraint family, and refinement-stable exact repair semantics. Then the local Euclidean repair rate depends only on active collar type, the active collar-type set is finite across the cofinal refinement family, and there is $c_* > 0$, independent of r , such that*

$$c_C \geq c_* \quad \text{for every active collar } C \in \mathcal{C}_r. \quad (3)$$

Proof. Active collars are exactly the collars on which complementary invisible fiber data are genuinely relaxed. Hence $D_C \neq 0$ on $\ker(E_C)$, so $c_C > 0$. Finite local combinatorial type gives a finite list of bounded collar patterns, including their visible interface alphabets, hidden fiber cardinalities, local constraint templates, and admissible repair maps. Branch homogeneity makes the conditioned MaxEnt relaxation scalar a function of this finite type data. Refinement stability says that refinement replaces a collar by copies of the same bounded type templates with the same normalized local Euclidean repair semantics, so no new rate-degenerating collar type appears along the cofinal family. Taking the minimum of c_C over the finite active type list gives $c_* > 0$. $\square$

8 Finite-Stage Gap

Proposition 8.1 (Special commuting-color finite-stage repair gap). *Assume that the collar family admits a bounded-color decomposition*

$$\mathcal{C}_r = \bigsqcup_{a=1}^q \mathcal{C}_{r,a}, \quad q < \infty, \quad (4)$$

independent of r , and that exact quotient-local gluing makes the corresponding color expectations commute. Then

$$L_r^{\text{rep}} \geq c_*(I - P_{0,r}), \quad (5)$$

where $P_{0,r}$ projects onto constants in K_r . Hence

$$\Delta_{\text{rep},r} \geq c_* > 0. \quad (6)$$

Proof. For each color a , define the parallel color expectation

$$E_{r,a} := \prod_{C \in \mathcal{C}_{r,a}} E_C.$$

Same-color collars are disjoint, so the factors commute. Exact quotient-local gluing makes $\{E_{r,a}\}_{a=1}^q$ a commuting family of orthogonal projections on the vacuum branch.

Repair completeness says that the only support-visible observables fixed by every local repair are constants:

$$\bigcap_{a=1}^q \text{Ran}(E_{r,a}) = \mathbb{C} \mathbf{1}_r. \quad (7)$$

Define

$$\tilde{L}_r := \sum_{a=1}^q c_*(I - E_{r,a}).$$

For commuting projections,

$$I - \prod_C E_C \leq \sum_C (I - E_C),$$

so $\tilde{L}_r \leq L_r^{\text{rep}}$. The projections $E_{r,a}$ are simultaneously diagonalizable. On the joint all-ones eigenspace, (7) says the vector is constant. Every nonconstant joint eigenspace has at least one color eigenvalue zero, so

$$\tilde{L}_r \geq c_*(I - P_{0,r}).$$

This proves (5), and the spectral gap bound (6) follows. $\square$

Remark 8.2 (Generic repair-gap certificate). The commuting-color proposition is a special certificate, not the generic repair-generator route. For overlapping collars one must instead compute

$$\kappa_r = \inf_{f \perp \mathbf{1}} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a lower bound $\gamma_* > 0$, then

$$L_r^{\text{rep}} \geq \gamma_* \kappa_r (I - P_{0,r}).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator; a uniform refinement lower bound is a separate receipt.

9 Continuum Extraction

Definition 9.1 (Finite compact-gauge state spaces and local cylinders). *At regulator r , make the following construction both for the finite four-dimensional Euclidean slab $\Lambda_r^{(4)} = (V_r^{(4)}, E_r^{(4)})$ and for its time-zero spatial slice $\Lambda_r^{(0)} = (V_r^{(0)}, E_r^{(0)})$. In the formulas below $\Lambda_r = (V_r, E_r)$ denotes either choice until the superscript is restored. Let D_r be the finite repaired-record and zero-obstruction sector register. In the gauge-register presentation, let*

$$\tilde{X}_r \subseteq G^{E_r} \times D_r$$

be the closed set obeying the finite Gauss, overlap, and repaired-record constraints. With $\mathcal{G}_r = G^{V_r}$ acting at edge endpoints and $\sim_{\text{ov}}$ the closed relation of support-visible overlap/presentation indistinguishability, define

$$X_r := (\tilde{X}_r / \mathcal{G}_r) / \sim_{\text{ov}}$$

as the support-visible quotient. This is compact Hausdorff. The regulator has finitely many cells, but X_r need not be a finite set when G is a compact Lie group.

For each active collar C , let

$$\rho_{C,r} : X_r \rightarrow Y_{C,r}$$

be the continuous complete repaired readback. The complementary conditional fiber contains every record or holonomy coordinate actually resampled by repair and is either finite uniform or a standard atomless probability space with its conditioned MaxEnt/Haar law. The readback retains exactly the repaired fixed datum, not every pre-repair holonomy. A quantum-link carrier enters this proposition only through its commuting Euclidean cylinder/readout subalgebra; the noncommutative carrier algebra would require a separate GNS-representation-kernel support reduction.

For a bounded cell region $O \subset \Lambda_r$, let $\mathfrak{C}_r^{G,\text{sv}}(O)$ be generated by the gauge-invariant Peter–Weyl matrix coefficients contracted into spin networks, Wilson-loop characters, and retained gauge-invariant boundary carriers in O , together with the overlap-sector projectors and continuous functions of the complete repaired collar readbacks in O . Define

$$\mathcal{A}_r^{G,\text{sv}}(O) := \overline{\mathfrak{C}_r^{G,\text{sv}}(O)}^{\|\cdot\|_\infty} \subseteq C(X_r).$$

Equivalently, start from the freely presented gauge-invariant readout-generator $*$ -algebra, quotient by the kernel of its evaluation homomorphism on X_r , and complete. This kernel is a closed two-sided overlap-trivial ideal chosen before a state and is distinct from the limiting GNS null ideal below. The relation $\sim_{\text{ov}}$ identifies exactly the configurations on which all declared spin-network, sector, and repaired-readback generators agree. The resulting self-adjoint unital algebra separates points of X_r , so Stone–Weierstrass gives

$$\mathcal{A}_r^{G,\text{sv}}(\Lambda_r) = C(X_r).$$

For any selected stage measure π_r , its GNS cylinder closure is therefore the full $L^2(X_r, \pi_r)$, after the finite π_r -null support reduction.

Write the two resulting systems as

$$(X_r^{(4)}, \mathcal{A}_r^{G,\text{sv},(4)}(O)) \quad \text{and} \quad (X_r^{(0)}, \mathcal{A}_{r,0}^{G,\text{sv}}(B)).$$

Euclidean-history restriction gives a continuous time-zero map $\tau_{0,r} : X_r^{(4)} \rightarrow X_r^{(0)}$.

For $r \preceq s$, coarse shadow multiplies the ordered fine-edge holonomies above each coarse edge and restricts the repaired records and sector labels. Gauge covariance and refinement associativity give

$$p_{sr} : X_s \rightarrow X_r, \quad p_{tr} = p_{sr} \circ p_{ts},$$

with the four-dimensional/time-zero commuting square

$$\tau_{0,r} \circ p_{sr}^{(4)} = p_{sr}^{(0)} \circ \tau_{0,s},$$

and unital local $*$ -maps on the separated support-visible quotient,

$$\iota_{rs}^O : \mathcal{A}_r^{G,\text{sv}}(O) \rightarrow \mathcal{A}_s^{G,\text{sv}}(O_s), \quad \iota_{rs}^O(a) = a \circ p_{sr}, \quad \iota_{rt}^O = \iota_{st}^O \iota_{rs}^O.$$

These maps preserve local isotony, gauge invariance, repaired readbacks, and the zero-obstruction sector. Their sector-projector restriction is the finite algebra injection in the compact paper's refinement/fiber-descent theorem [2]. We use a countable cofinal regulator sequence and countable bounded-region exhaustion. The unrestricted directed version uses a subnet and product compactness instead. If every declared coarse configuration has a fine extension, p_{sr} is surjective and the full pull-back is injective. Otherwise the C^* -inductive limit automatically quotients the common refinement-null kernel; Proposition 9.2 does not require prelimit injectivity. Write $\mathcal{A}_r^{G,sv} := \mathcal{A}_r^{G,sv}(\Lambda_r)$.

Proposition 9.2 (Projective weak- $*$ extraction and support-visible GNS gluing). *Let $\mu_s^{(4)}$ be the declared finite quotient-Gibbs probability measure on $X_s^{(4)}$, let*

$$\pi_s := (\tau_{0,s})_{\#} \mu_s^{(4)}, \quad \mu_s^{(0)} := \pi_s,$$

and, for $d \in \{4, 0\}$, define

$$\omega_s^{(d)}(a) := \int_{X_s^{(d)}} a d\mu_s^{(d)}, \quad \omega_{s \downarrow r}^{(d)} := \omega_s^{(d)} \circ \iota_{rs}^{(d)}.$$

The following statements hold for both values of d ; the superscript is suppressed in the proof.

(i) for $q \preceq r \preceq s$,

$$\omega_{s \downarrow r} \circ \iota_{qr} = \omega_{s \downarrow q};$$

(ii) a cofinal subsequence, or subnet, has weak- $*$ limits $\omega_{s_\alpha \downarrow r} \rightarrow \bar{\omega}_r$ on every fixed local cylinder, and

$$\bar{\omega}_r = \bar{\omega}_t \circ \iota_{rt} \quad (r \preceq t);$$

(iii) the local limits glue to a state ω_∞ on

$$\mathcal{A}_\infty^{G,cyl}(O) := \overline{\lim_{\rightarrow} \mathcal{A}_r^{G,sv}(O_r)},$$

and the union of these isotone local algebras is dense in the global cylinder algebra;

(iv) after the separate support reduction

$$\mathcal{N}_\omega(O) := \{a \in \mathcal{A}_\infty^{G,cyl}(O) : \omega_\infty(a^*a) = 0\}, \quad \mathcal{A}_\infty^{G,supp}(O) := \mathcal{A}_\infty^{G,cyl}(O) / \mathcal{N}_\omega(O),$$

the induced state and representation are faithful. The compatible local GNS spaces have a Hilbert direct limit $(K^{(d)}, \Pi^{(d)}, \mathbf{1}^{(d)})$, and the union of their cylinder images is dense. Write $K^E := K^{(4)}$ for the Euclidean-history GNS space and $K := K^{(0)}$ for the time-zero repair GNS space. The latter vector becomes the physical vacuum only through the transfer/OS certificate.

If, in a finite quotient presentation, the four-dimensional measures have declared weights

$$\mu_r^{(4)}(x) = Z_r^{-1} m_r(x) e^{-S_r(x)}$$

and satisfy the fiber-sum identity

$$\sum_{x' : p_{sr}(x')=x} m_s(x') e^{-S_s(x')} = \alpha_{sr} m_r(x) e^{-S_r(x)} \quad (\text{FS})$$

with α_{sr} independent of x , then $(p_{sr})_{\#} \mu_s^{(4)} = \mu_r^{(4)}$. The time-zero square gives $(p_{sr})_{\#} \pi_s = \pi_r$, so both original state families are projective. Without (FS), compactness produces a projective cluster family; it does not prove $\bar{\omega}_r = \omega_r$ or uniqueness of the continuum state. For continuous compact-holonomy stages, replace (FS) by the corresponding disintegration receipt $(p_{sr})_{\#} \mu_s^{(4)} = \mu_r^{(4)}$.

Proof. For $q \preceq r \preceq s$, coarse-shadow functoriality gives

$$(\omega_s \circ \iota_{rs}) \circ \iota_{qr} = \omega_s \circ \iota_{qs},$$

proving (i).

For a fixed local cylinder algebra, its state space is weak-* compact by Banach–Alaoglu. The chosen local algebras are separable because compact metrizable G has a countable Peter–Weyl test algebra and a finite regulator has only finitely many record and sector generators. Their state spaces are therefore weak-* metrizable. Successively extract a subsequence for the first, second, and later cylinders, choosing the k -th diagonal index beyond the k -th regulator stage. The diagonal subsequence is cofinal and converges on all cylinders. For a general directed regulator set, use a convergent subnet in the compact product of the local state spaces; unresolved early coordinates may be filled arbitrarily because each fixed coordinate is genuinely resolved on a cofinal tail. Weak-* continuity of precomposition and part (i) yield

$$\bar{\omega}_r \circ \iota_{qr} = \lim_{\alpha} \omega_{s_{\alpha} \downarrow r} \circ \iota_{qr} = \lim_{\alpha} \omega_{s_{\alpha} \downarrow q} = \bar{\omega}_q,$$

which proves (ii).

For a representative $[a]_r$ in the algebraic inductive limit, define

$$\omega_{\infty}([a]_r) := \bar{\omega}_r(a).$$

Compatibility makes this well defined. Positivity and normalization are checked in one finite algebra containing the element. For every $t \succeq r$,

$$|\bar{\omega}_r(a)| = |\bar{\omega}_t(\iota_{rt}(a))| \leq \|\iota_{rt}(a)\|.$$

Taking the infimum along the tail gives the inductive-limit norm bound even for noninjective bonding maps, so the state extends uniquely to the C^* -completion. This proves (iii).

On a commutative local cylinder algebra, the Riesz representation theorem identifies the null ideal with the continuous functions that vanish on the measure support. The support quotient is therefore $C(\text{supp } \mu_O)$, where μ_O is the representing local Radon measure, and its induced state is faithful. If $O \subseteq O'$, state compatibility gives

$$a \in \mathcal{N}_{\omega}(O) \iff \iota_{OO'}(a) \in \mathcal{N}_{\omega}(O').$$

Thus the inclusions descend injectively and the support quotients form an isotone faithful local net. For $r \preceq t$, define

$$V_{rt}[a]_r := [\iota_{rt}(a)]_t.$$

It is well defined and isometric because

$$\|V_{rt}[a]_r\|^2 = \bar{\omega}_t(\iota_{rt}(a^*a)) = \bar{\omega}_r(a^*a).$$

These maps compose, preserve $\mathbf{1}$, and intertwine the cylinder representations. Their Hilbert direct limit is the GNS completion of the cylinder union, proving (iv).

Finally, suppressing the $d = 4$ superscript, summing (FS) over x gives $Z_s = \alpha_{sr} Z_r$. Division by the partition functions then gives $(p_{sr}^{(4)})_{\#} \mu_s^{(4)} = \mu_r^{(4)}$, equivalently $\omega_s^{(4)} \circ \iota_{rs}^{(4)} = \omega_r^{(4)}$. The commuting time-zero square gives the corresponding $d = 0$ identity. $\square$

Remark 9.3 (What compactness does not preserve). The overlap-trivial presentation quotient and the GNS support quotient are different. Even faithful finite states can converge to a nonfaithful state. Proposition 9.2 constructs the cylinder algebra, its state, the repair-side GNS space K , and the constant vacuum vector. It does not construct $\mathcal{H}$, a limit generator, a physical intertwiner, or vacuum-projection convergence. Those require the finite dynamic receipts below.

Assumption 9.4 (Four-dimensional Yang–Mills continuum certificate). *In addition to the finite compact-gauge repair data, one cofinal regulator family supplies all of the following.*

- (Y1) *For each bounded region O , renormalized gauge-invariant cylinder maps from one fixed countable test $*$ -algebra $\mathfrak{C}_0^G(O)$ to $\mathcal{A}_r^{G,\text{sv}}(O_r)$, with a uniform finite-volume/lattice-spacing Cauchy bound. The maps commute with the coarse-shadow ι_{rs} up to those vanishing Cauchy defects. This makes the extracted cluster family unique on the declared tests.*
- (Y2) *Exact finite reflection positivity, or a positive self-adjoint transfer matrix, with reflection-compatible renormalization maps.*
- (Y3) *Uniform restoration of Euclidean covariance, locality, regularity, clustering, and the declared vacuum-sector condition.*
- (Y4) *Generalized Mosco convergence of both the physical transfer forms q_r^{phys} and repair forms q_r^{rep} to closed nonnegative continuum forms, or exact coherent direct-limit semigroup identities under state-preserving repair and physical refinement isometries, with one refinement-independent physical Euclidean-time normalization. In the exact alternative the isometries send $\mathbf{1}_r$ to $\mathbf{1}_s$ and Ω_r to Ω_s .*
- (Y5) *Convergence of the finite intertwiners U_r and their inverses on dense form cores to a unitary U , together with $U_r\Omega_r = \mathbf{1}_r$ and strong convergence of $P_{0,r} = |\mathbf{1}_r\rangle\langle\mathbf{1}_r|$ to $P_0^K = |\mathbf{1}\rangle\langle\mathbf{1}|$ under the declared Hilbert identifications. The physical vacuum projection is $P_0^{\mathcal{H}} := U^{-1}P_0^K U = |\Omega\rangle\langle\Omega|$. On the dense time-zero cylinder core, the finite Markov/OS identity*

$$\omega_r^{(4)}\left((\iota_{0,r}f)^*\tau_t^{(r)}(\iota_{0,r}g)\right) = \langle f, e^{-tL_r^{\text{rep}}}g \rangle_{K_r} = \langle U_r^{-1}f, e^{-tH_r}U_r^{-1}g \rangle_{\mathcal{H}_r}$$

holds and passes to the limit. Thus the OS-reconstructed time-zero Hilbert space and Hamiltonian are the same $(\mathcal{H}, H)$ as the transfer-form limit, not a second pair with reused notation.

- (Y6) *A nontriviality certificate: renormalized local observables A_r , all images of one fixed bounded cylinder test under the maps in (Y1), whose centered vectors converge under the Hilbert identifications and satisfy*

$$\inf_r \text{Var}_{\omega_r}(A_r) > 0, \quad \sup_r q_r^{\text{phys}}(A_r\Omega_r) < \infty.$$

The explicit multiresolution construction in the main paper supplies a model for items (Y1)–(Y2) and an exact transfer tower when its shell identities are implemented. Identification of its continuum limit with the four-dimensional Yang–Mills OS theory is the additional content of (Y3)–(Y6).

Theorem 9.5 (Conditional continuum transfer identification). *Under Assumption 9.4, the generalized Mosco limit, or the exact coherent direct limit in item (Y4), produces nonnegative self-adjoint generators H and L^{rep} and a unitary U satisfying*

$$Ue^{-tH}U^{-1} = e^{-tL^{\text{rep}}} \quad (t \geq 0), \quad (8)$$

and hence

$$UHU^{-1} = L^{\text{rep}}. \quad (9)$$

If the finite repair generators satisfy

$$L_r^{\text{rep}} \geq c_*(I - P_{0,r})$$

with one $c_* > 0$ and the vacuum projections converge strongly, then

$$H \geq c_*(I - P_0^{\mathcal{H}}).$$

Proof. Generalized Mosco convergence is equivalent to generalized strong-resolvent and transfer-semigroup convergence on the declared Hilbert identifications. In the exact alternative, the coherent finite contraction semigroups define strongly continuous direct-limit semigroups on the dense cylinder images, and uniform contractivity extends them to the completed Hilbert spaces. Their bounded operators are symmetric and positive on the dense finite-stage union, hence self-adjoint and positive after extension. Convergence of U_r and U_r^{-1} on dense form cores passes the finite-stage intertwining relation to the limit, so uniqueness of self-adjoint semigroup generators gives $UHU^{-1} = L^{\text{rep}}$. For a repair-form Mosco recovery sequence $\psi_r \rightarrow \psi$,

$$q_r^{\text{rep}}(\psi_r) \geq c_*(\|\psi_r\|^2 - \|P_{0,r}\psi_r\|^2).$$

Taking the limit and using strong convergence of the vacuum projections gives the continuum form inequality. In the exact direct-limit alternative, the finite bound is equivalently

$$\|e^{-tL_r^{\text{rep}}}(I - P_{0,r})\| \leq e^{-c_*t};$$

coherence passes this estimate to the direct-limit semigroup, and the spectral theorem gives the same continuum form inequality. $\square$

10 Axiomatic Reconstruction and Nontriviality

Theorem 10.1 (Conditional Osterwalder–Schrader reconstruction on the support-visible compact-gauge branch). *Under Assumptions 5.1 and 9.4, the continuum support-visible compact-gauge cylinder family is Euclidean invariant, reflection positive, regular on gauge-invariant local cylinder observables, nontrivial, and cyclic for the vacuum sector. Hence Osterwalder–Schrader reconstruction gives a four-dimensional quantum Yang–Mills theory*

$$(\mathcal{H}, \Omega, H, \mathcal{A}_{\text{loc}}^G)$$

on the support-visible gauge-invariant local algebra, with $H \geq 0$ and e^{-tH} equal to the certified continuum Euclidean transfer semigroup.

Proof. Euclidean covariance, locality, reflection positivity, regularity, clustering, transfer convergence, and nontriviality are precisely the certificate items in Assumption 9.4. Proposition 9.2 supplies the four-dimensional Euclidean cylinder state K^E and its time-zero GNS restriction K ; the certificate supplies the missing regularity and transfer controls. The finite/limiting time-zero identity in (Y5) identifies the OS time-zero inner product and translation semigroup with the transfer-form limit $(\mathcal{H}, H)$. The standard Osterwalder–Schrader reconstruction theorem therefore produces the Hilbert space, vacuum, local algebra, and positive Hamiltonian, and identifies the physical time-translation semigroup with the certified Euclidean transfer semigroup [10, 11]. $\square$

Proposition 10.2 (Nontriviality of the support-visible compact-gauge theory). *Under Assumption 9.4, the support-visible compact-gauge local algebra on the zero-obstruction vacuum branch strictly contains the vacuum scalars and admits a non-vacuum finite-energy local excitation.*

Proof. The compact-gauge witness and physical-UV landing theorem in the OPH compact paper supplies a realized nontrivial compact-gauge branch [2]. At finite cutoff this gives a support-visible gauge-invariant local observable, such as a nonconstant Wilson/plaquette cylinder observable, whose vacuum variance is positive. Its GNS vector is orthogonal to the vacuum after subtracting its expectation value. The finite-range local-Gibbs generator assigns finite energy to finite-cylinder excitations. The continuum nontriviality step is not a consequence of weak-* state convergence alone; it is the variance-floor and finite-energy item (Y6) in Assumption 9.4. With that certificate, the local cylinder vectors survive in the support-visible continuum, so the continuum local algebra is strictly larger than $\mathbb{C}I$ and contains non-vacuum finite-energy local excitations. $\square$

11 Main Theorem

Theorem 11.1 (Conditional positive four-dimensional compact-gauge Yang–Mills mass gap). *Let G be a compact simple gauge group carried by a support-visible compact-gauge OPH vacuum branch satisfying the standing setup above and the continuum certificate of Assumption 9.4, the finite transfer receipt of Assumption 7.3, and all hypotheses of Theorem 7.4, Lemma 7.5, and Proposition 8.1. The theory reconstructed in Theorem 10.1 is nontrivial, and its continuum support-visible Hamiltonian H satisfies*

$$H \geq c_*(I - P_0^{\mathcal{H}}), \quad (10)$$

where $P_0^{\mathcal{H}}$ projects onto the physical vacuum. Therefore

$$\text{Spec}(H) \cap (0, c_*) = \emptyset, \quad \Delta_{\text{YM}} \geq c_* > 0. \quad (11)$$

On that certified continuum branch the repair gap and Yang–Mills gap are exactly equal:

$$\Delta_{\text{YM}} = \Delta_{\text{rep}}. \quad (12)$$

Proof. At every finite stage, Proposition 8.1 gives

$$L_r^{\text{rep}} \geq c_*(I - P_{0,r}).$$

By Theorem 9.5, on the extracted cylinder/GNS space, this lower bound passes to the support-visible continuum:

$$L^{\text{rep}} \geq c_*(I - P_0^K).$$

Using (9), $UHU^{-1} = L^{\text{rep}}$. Conjugating the lower bound by U^{-1} gives (10), and the spectral statement (11) follows. Since unitary equivalence preserves the nonzero spectrum and identifies the vacuum with the constant sector, the infimum of the nonzero spectrum is the same on both sides, giving (12). $\square$

Remark 11.2 (Group-uniform form). The proof is group-uniform. Once a compact simple G is carried by a compact-gauge zero-obstruction OPH branch, no step uses special properties of the realized Standard Model quotient. The inputs are compact-gauge support-visible quotient locality, exact local repair on collars, the finite ground-state-transform/cross-fiber receipt, the uniform rate and commuting-color receipts, refinement coherence, projective cylinder extraction, the transfer/vacuum certificate, and the OS regularity/noncollapse certificate.

12 Exact Gap Accounting

On a branch satisfying Assumption 9.4, the proof gives more than a positive lower bound. It identifies the Hamiltonian whose gap is being measured:

$$H = U^{-1} L^{\text{rep}} U.$$

Therefore

$$\text{Spec}(H) \setminus \{0\} = \text{Spec}(L^{\text{rep}}) \setminus \{0\}.$$

The Yang–Mills gap is exactly the first nonzero repair eigenvalue:

$$\Delta_{\text{YM}} := \inf(\text{Spec}(H) \setminus \{0\}) = \inf(\text{Spec}(L^{\text{rep}}) \setminus \{0\}) =: \Delta_{\text{rep}}.$$

The finite-stage color argument proves $\Delta_{\text{rep}} \geq c_* > 0$. The exact accounting statement is the conditional equality $\Delta_{\text{YM}} = \Delta_{\text{rep}}$; the inequality is the positivity proof for that same quantity. Without the continuum certificate, the same equations remain finite-regulator repair accounting instead of a Clay-admissible Yang–Mills theorem.

13 Relation to the 2D Yang–Mills Bridge

The OPH corpus also contains an exact edge-sector identity:

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1),$$

which is the compact-group heat kernel at the identity and matches the standard two-dimensional Yang–Mills heat-kernel partition form. Peter–Weyl supplies the heat-kernel identity, and Gross–Taylor gives the standard large- N worldsheet dictionary when a separate large- N_{edge} branch with remainder control is declared [12, 13].

That 2D result is a normalization and worldsheet-effective bridge. The spectral theorem above concerns the support-visible compact-gauge Hamiltonian and obtains its lower bound by identifying Euclidean transfer with the repair generator. The string-vacuum selector uses the same bridge only as effective edge-language input; its critical-worldsheet, Bouchard–Donagi, safety-layer, threshold, and moduli-locking gates do not alter the four-dimensional compact-gauge mass-gap theorem surface [6].

14 Clay Deliverables Checklist

Clay/Jaffe–Witten target	OPH repair-dynamics boundary
Compact simple gauge group G	G is arbitrary compact simple, provided it is carried by the OPH compact-gauge zero-obstruction branch.

Clay/Jaffe–Witten target	OPH repair-dynamics boundary
Four-dimensional Euclidean Yang–Mills form	Theorem 5.2, imported from the compact paper’s D7 Yang–Mills theorem surface, gives $S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle d^4x$ under the branch and renormalized identification certificates.
Four-dimensional quantum Yang–Mills theory	Proposition 9.2 proves support-visible projective weak-* / GNS extraction. Renormalized Schwinger convergence, reflection positivity, Euclidean covariance/locality, clustering, nontriviality, and transfer/intertwiner convergence remain conditional on Assumption 9.4.
Nontriviality	Conditional on the variance-floor and finite-energy certificate in item (Y6). Finite nonconstant Wilson/plaquette cylinders are candidates; weak-* convergence alone is not enough.
Axiomatic strength	Theorem 10.1 states the conditional OS reconstruction step on the support-visible gauge-invariant local algebra.
Mass gap	Under the finite ground-state-transform, cross-fiber-rate, commuting-color, and uniform-floor receipts, $L_r^{\text{rep}} \geq c_*(I - P_{0,r})$. The continuum Hamiltonian bound $H \geq c_*(I - P_0^{\mathcal{H}})$ follows only after Theorem 9.5.
Exact gap accounting	On the certified continuum branch, $UHU^{-1} = L^{\text{rep}}$ gives $\Delta_{\text{YM}} = \Delta_{\text{rep}}$. OPH accounts exactly for the Yang–Mills gap as the first nonzero repair eigenvalue only on that branch.
Uniformity in G	The proof uses compact simplicity and compact-gauge quotient locality, not Standard-Model-specific representation data.

15 Conclusion

The OPH finite repair mechanism for the Yang–Mills mass gap is simple. A non-vacuum support-visible compact-gauge excitation is not fixed by all local repair collars. At least one active collar must relax it. Because active collar rates have a uniform positive floor and the collars can be organized into finitely many commuting colors, every non-vacuum finite-regulator state pays a positive Euclidean repair cost.

The Clay-facing conclusion is conditional. If the regulator family also satisfies the renormalized continuum, reflection-positivity, OS, nontriviality, and transfer/intertwiner certificate of Assumption 9.4, then the continuum compact-gauge Hamiltonian is unitarily equivalent to the repair generator. On that certified branch the Yang–Mills gap is the repair gap:

$$\boxed{\Delta_{\text{YM}} = \Delta_{\text{rep}} \geq c_* > 0.}$$

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